

Microstrip Patch Antenna Calculations

The single element patch antenna parameters are calculated analytically based on standard microstrip transmission line equations at an operational target frequency of $f_0 = 2.0$ GHz on a standard FR-4 substrate ($\epsilon_r = 4.3$, $h = 1.6$ mm, thickness $t = 0.035$ mm).

Width of the Patch (W_p)

The width controls the input impedance, radiating edge field distributions, and radiation pattern properties:

$$W_p = (c / 2f_0) * \sqrt{2 / (\epsilon_r + 1)}$$

Where $c = 3 \times 10^8$ m/s and $\epsilon_r = 4.3$. Substituting the explicit parameters yields:

$$W_p = (3 \times 10^8) / (2 \times (2 \times 10^9)) * \sqrt{2 / (4.3 + 1)} = 0.075 * \sqrt{0.3773} \approx 46.07 \text{ mm}$$

Effective Dielectric Constant (ϵ_{re})

To account for the fringing fields traveling partly in the substrate and partly in the free space air, the effective value is calculated via:

$$\epsilon_{re} = (\epsilon_r + 1) / 2 + ((\epsilon_r - 1) / 2) * [1 + 12(h / W_p)]^{-1/2}$$
$$\epsilon_{re} = ((4.3 + 1) / 2) + ((4.3 - 1) / 2) * [1 + 12(1.6 / 46.07)]^{-1/2} \approx 4.036$$

Fringing Extension Length (ΔL)

Due to the fringing fields distributed along the patch boundaries, the antenna looks electrically slightly longer than its absolute physical boundary dimensions:

$$\Delta L = 0.412 * h * [(\epsilon_{re} + 0.3)(W_p/h + 0.264)] / (\epsilon_{re} - 0.258)(W_p/h + 0.8)$$

Substituting the height $h = 1.6$ mm and width ratio terms yields an analytical extension value:

$$\Delta L \approx 0.73 \text{ mm}$$

Length of the Patch (L_p)

The physical resonant length of the patch element determines the core fundamental tracking operating band window and is derived as:

$$L_p = [c / (2f_0 * \sqrt{\epsilon_{re}})] \Delta L$$

$$L_p = [3 \times 10^8 / (2 \times 2.0 \times 10^9 * \sqrt{4.036})] - 2(0.73 \text{ mm}) \approx 35.85 \text{ mm}$$

Note: Parametric software modeling in CST sweeps and refines this physical baseline value down to 34.7 mm for ideal match conditions, and further optimizes it to 32.4 mm under $R = 300$ mm conformal bend stresses to fully neutralize feed pin loading shifts.

Single Element:

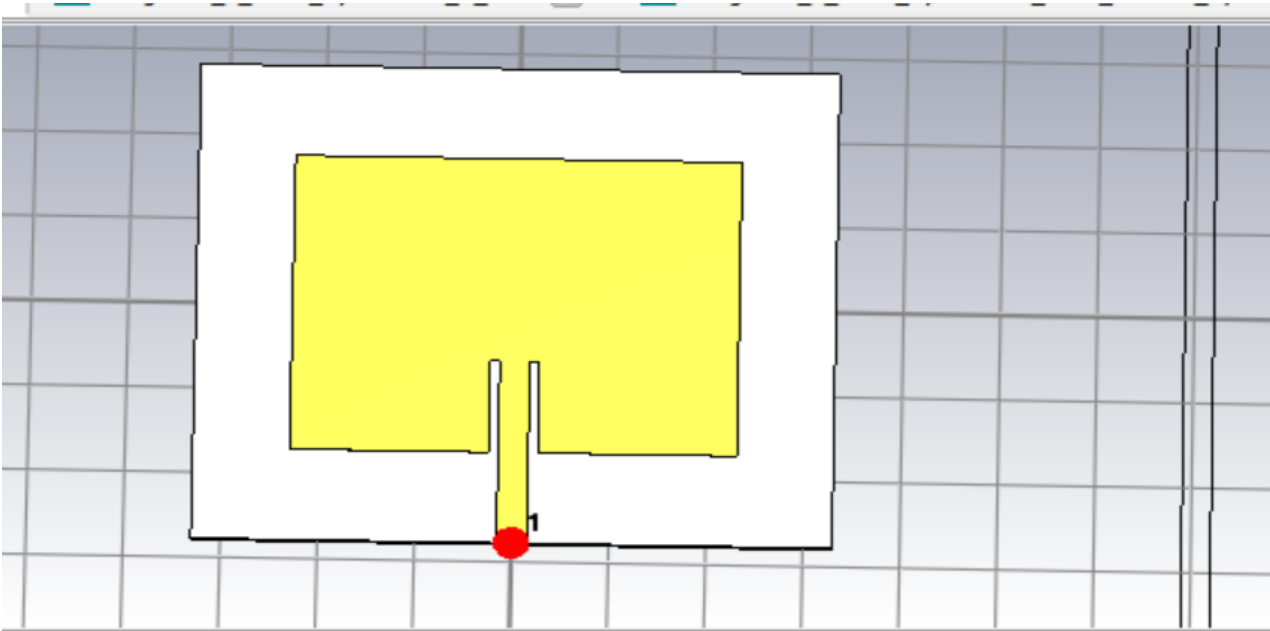
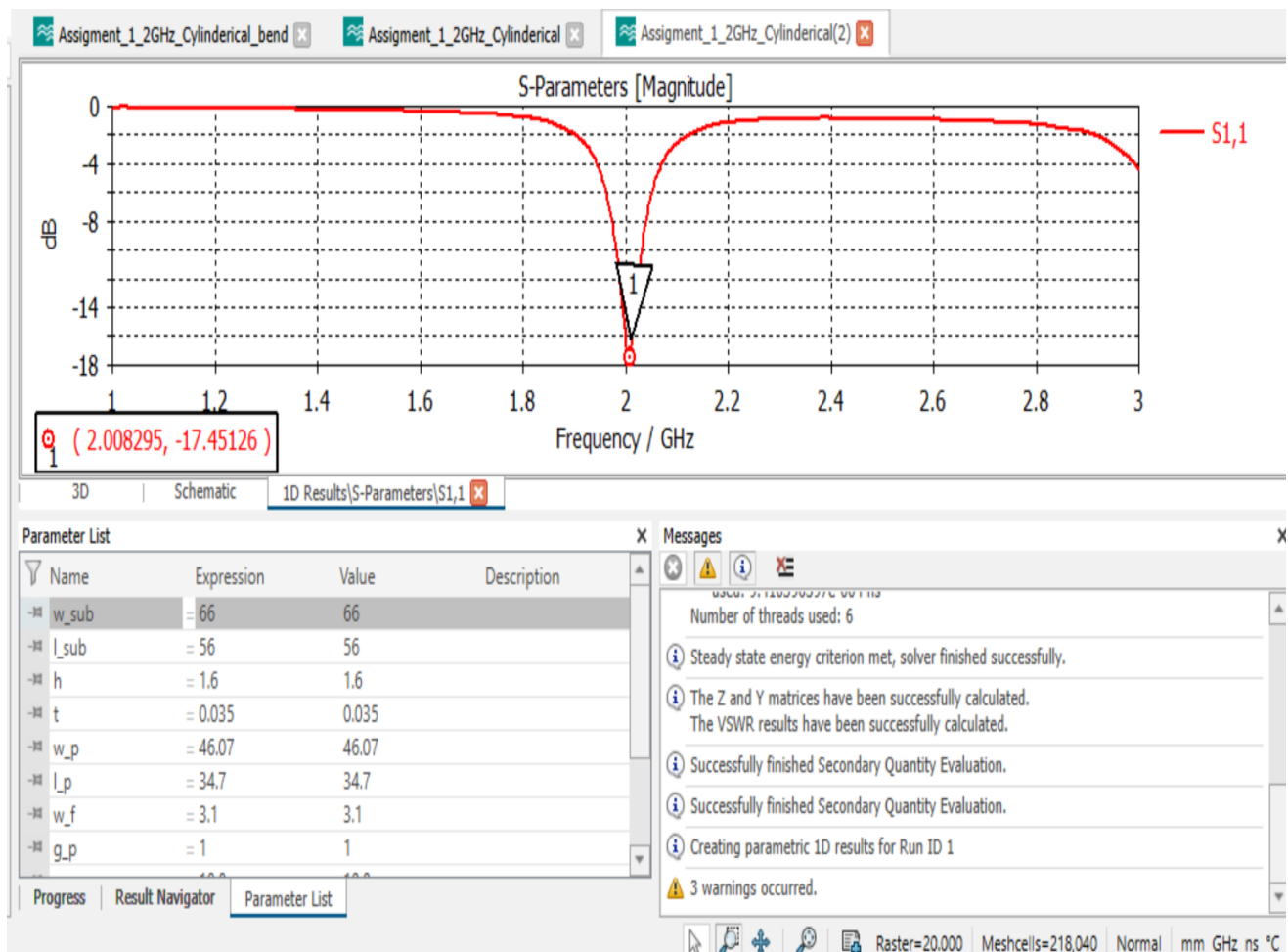
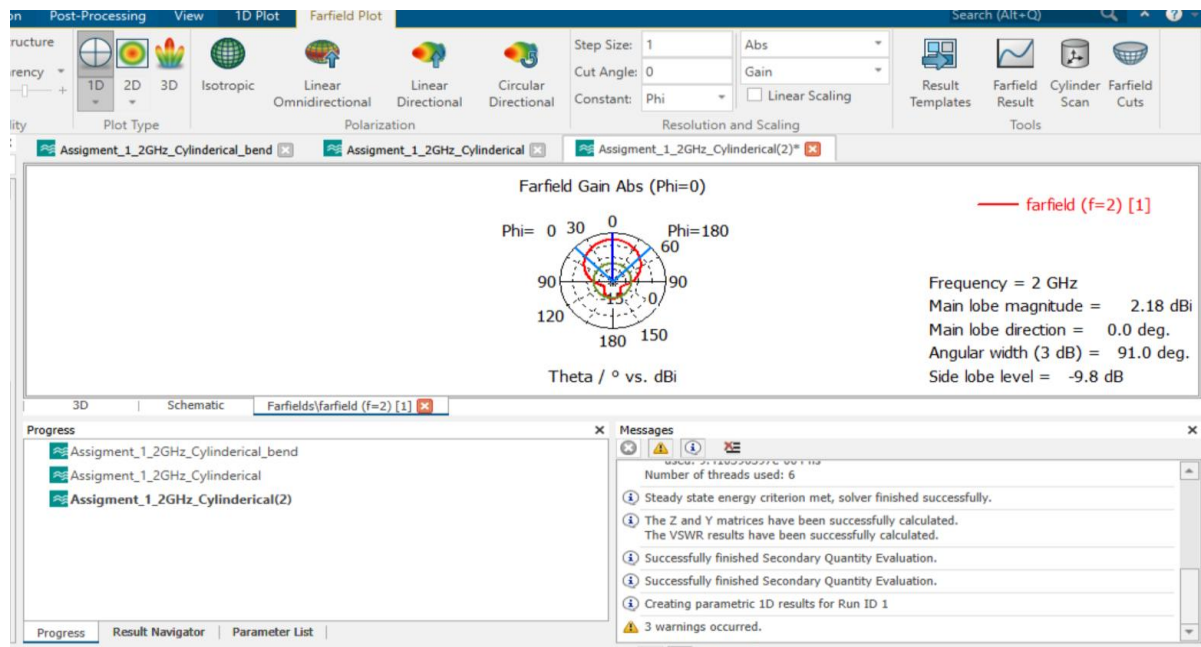


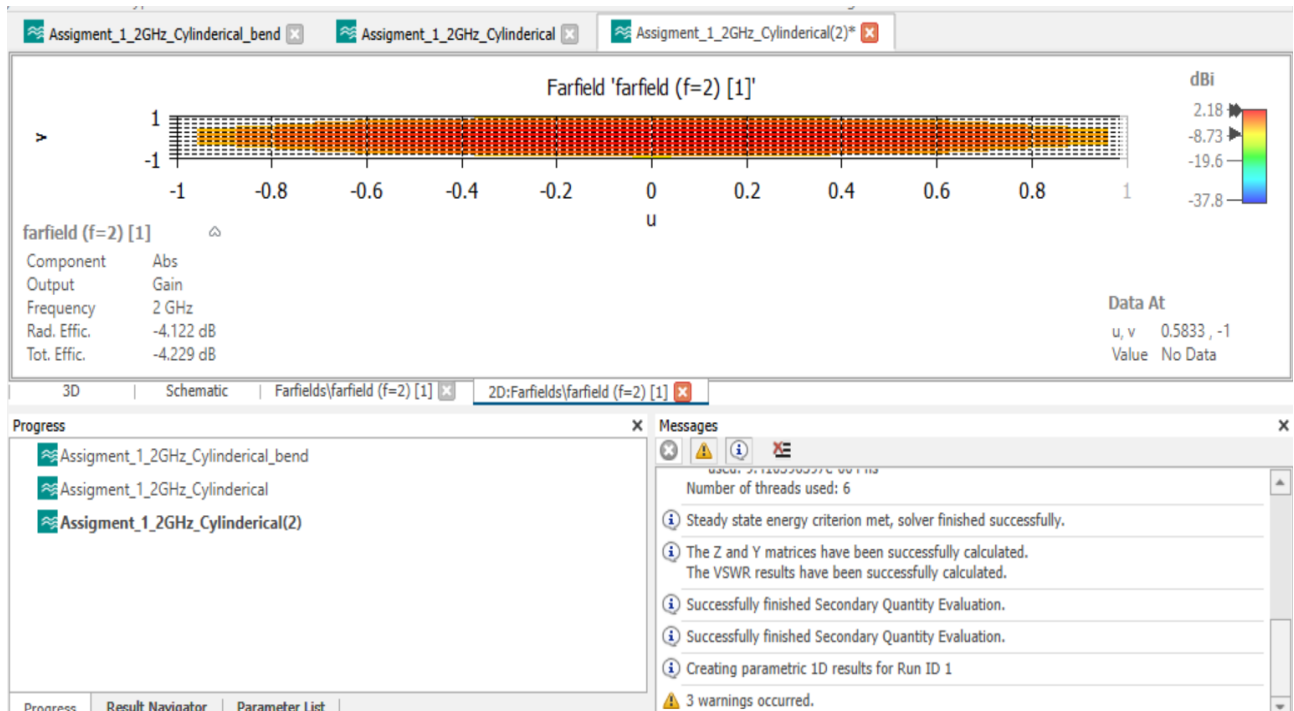
Figure 1: Single element in CST using Calculated data



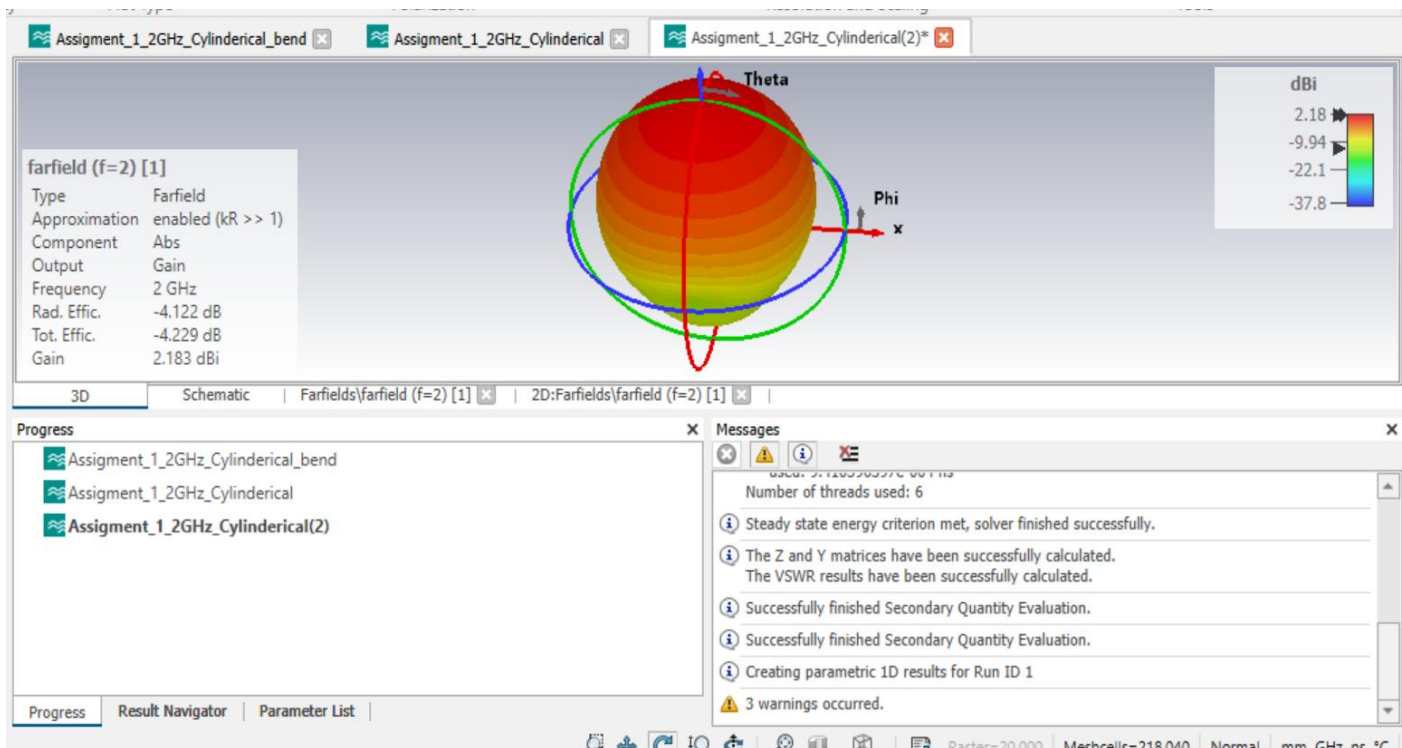
Farfeild Single Element:



Farfield 2D:



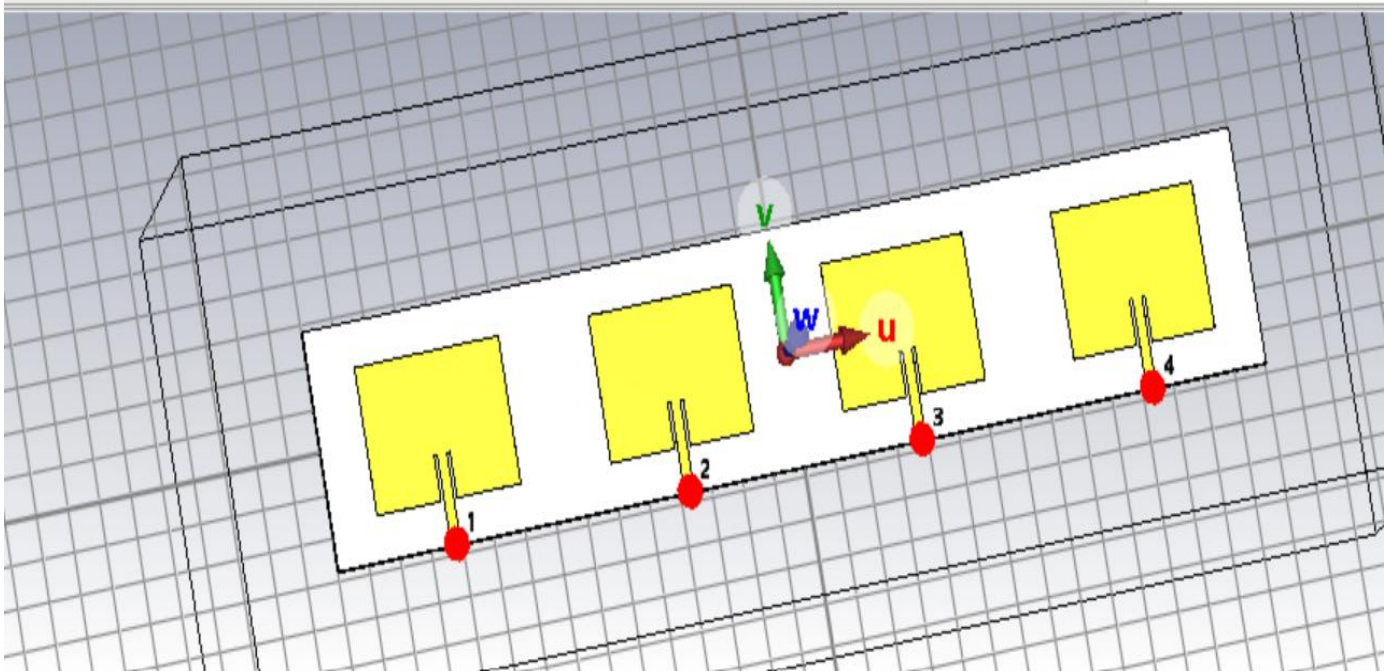
3D:



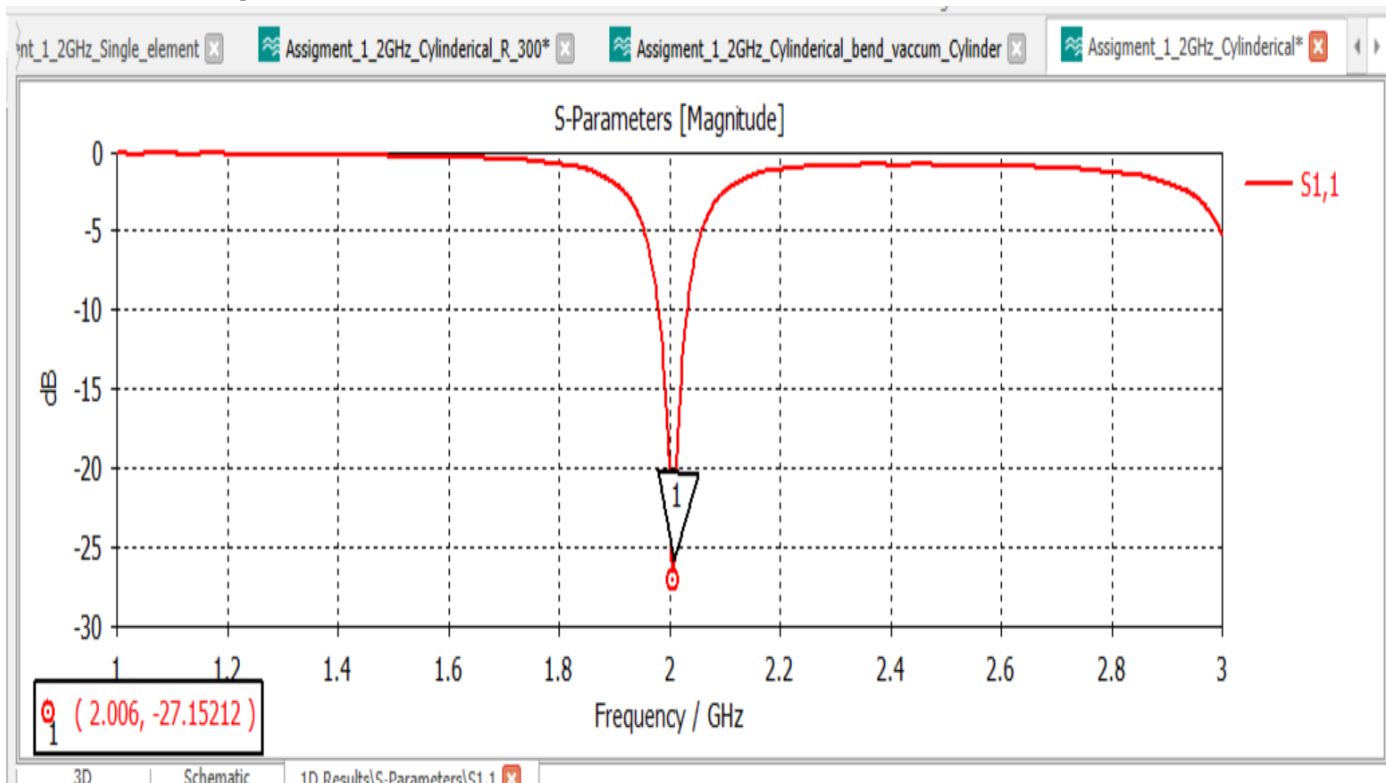
1X4 Linear Array: Calculation:

$$\lambda_0 = \frac{c}{f_0} = \frac{3 \times 10^8 \text{ m/s}}{2 \times 10^9 \text{ Hz}} = 0.15 \text{ m} = 150 \text{ mm}$$

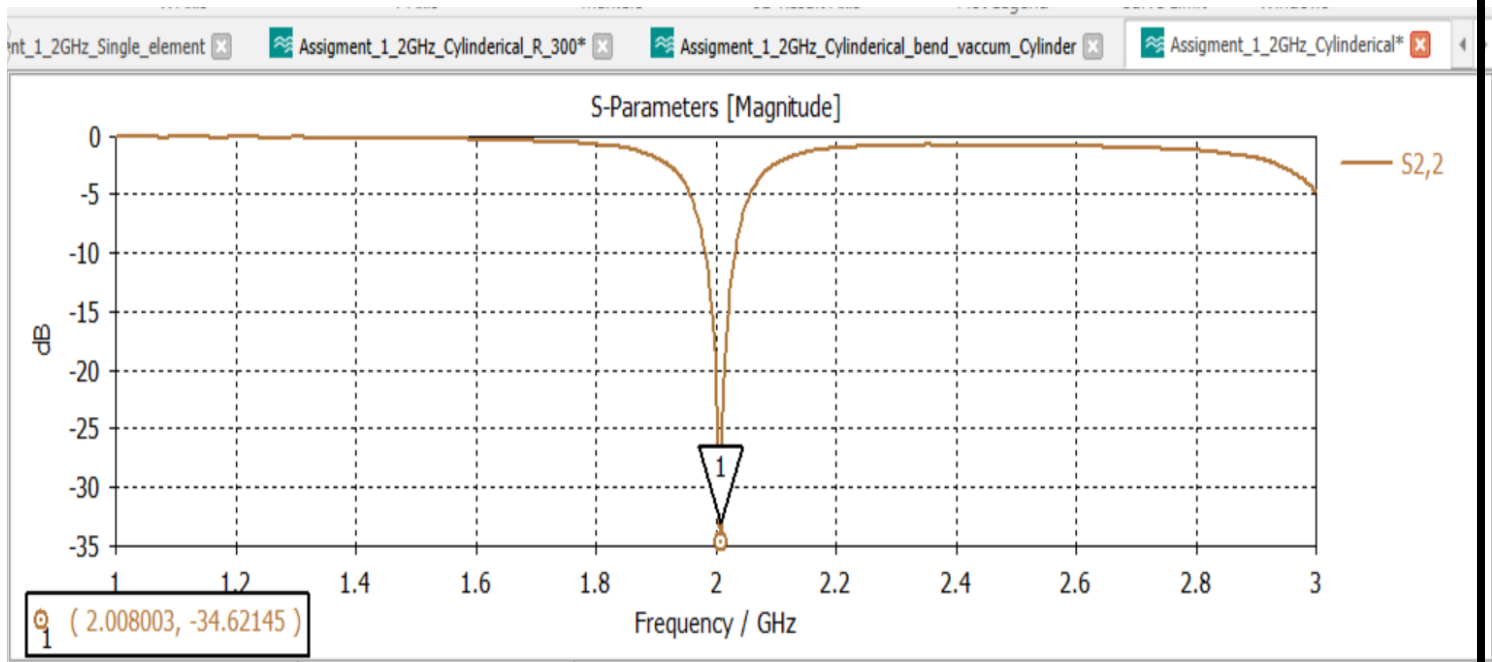
$$d = \frac{\lambda_0}{2} = \frac{150 \text{ mm}}{2} = \mathbf{75 \text{ mm}}$$



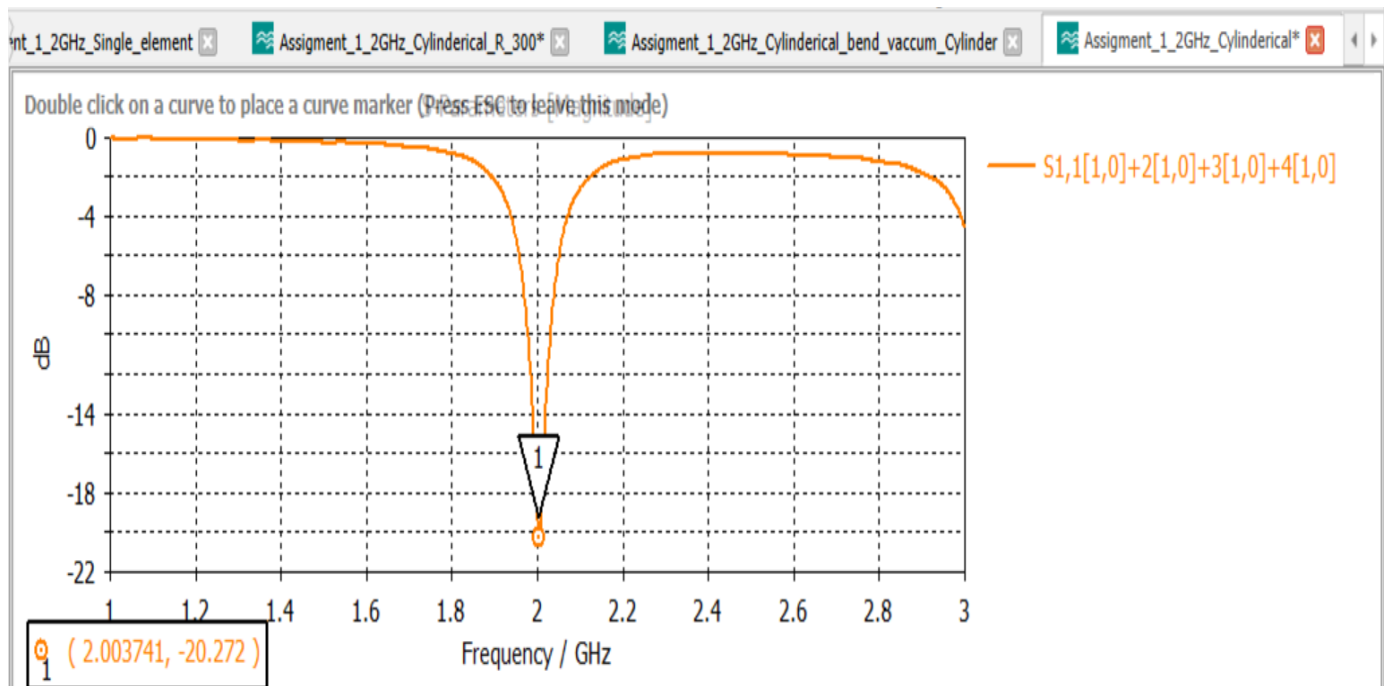
S11 Array:



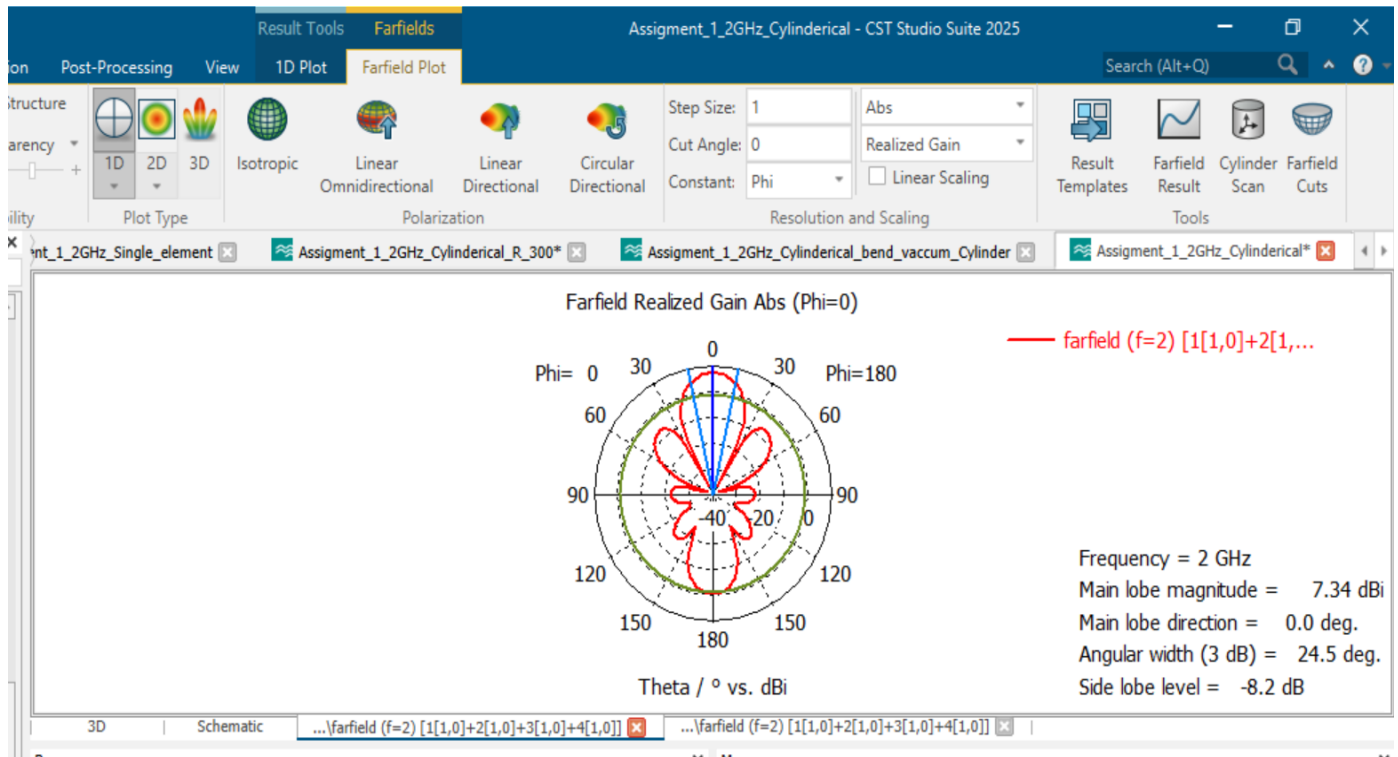
S22 Array:



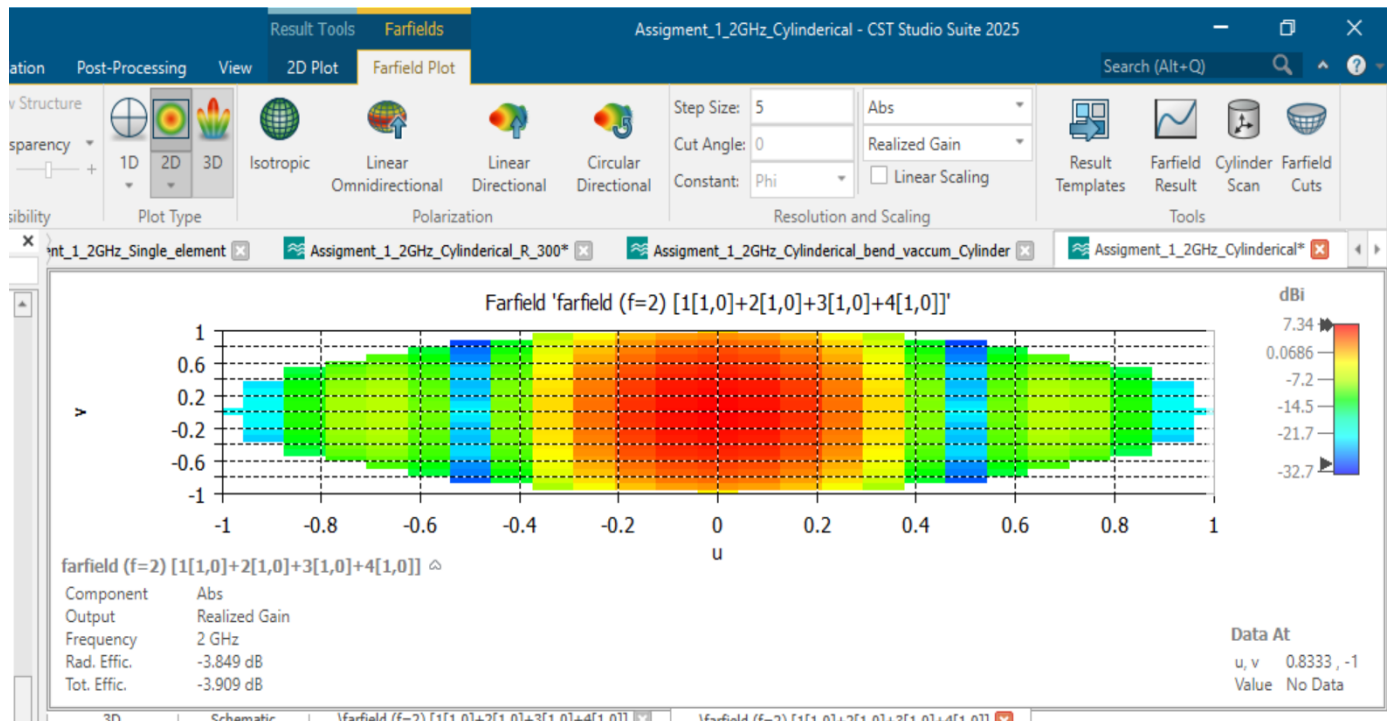
Combine :



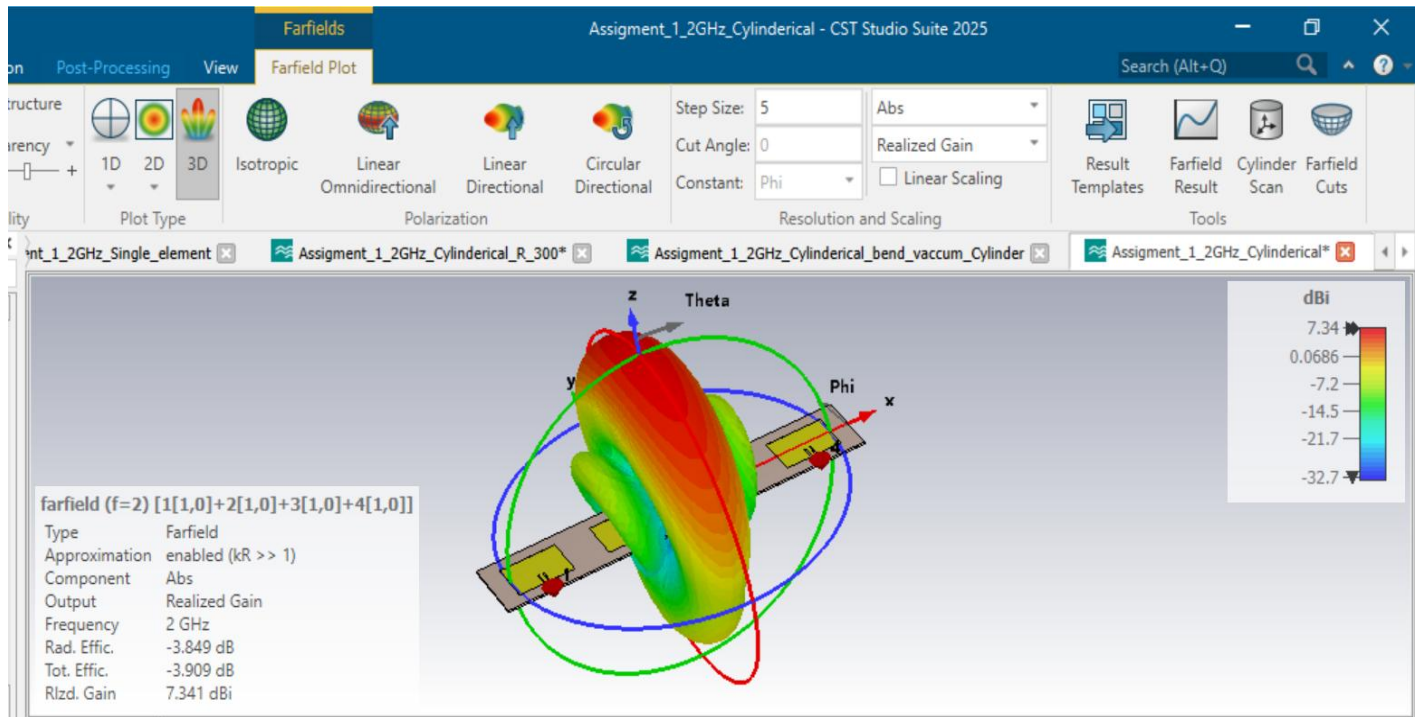
Farfeild 1D Array:



2D Array:



3D array:



Cylindrical Array R = 30cm:

1.1 Free-Space Wavelength Calculation

The operational frequency is given as $f = 2.0$ GHz. The free-space wavelength (λ_0) is calculated using the constant speed of light ($c \approx 3 \times 10^8$ m/s):

$$\lambda_0 = c / f = (3 \times 10^8 \text{ m/s}) / (2.0 \times 10^9 \text{ Hz}) = 0.15 \text{ m} = 150 \text{ mm}$$

1.2 Element Spacing

To suppress grating lobes while maximizing effective aperture area, the nominal element spacing (d) is established at exactly half the free-space wavelength ($0.5\lambda_0$):

$$d = \lambda_0 / 2 = 150 \text{ mm} / 2 = 75 \text{ mm}$$

2. Conformal Geometric Distortion & Broadside Phase Correction

Bending a linear antenna array onto a cylindrical surface of radius $R = 300$ mm introduces a physical displacement along the propagation axis. This geometric curvature causes an intrinsic phase lag in the outer radiating elements relative to the inner core elements, severely degrading the coherent broadside wavefront shape.

2.1 Angular Positioning on Cylinder Surface (α)

The physical arc length separating adjacent patch elements is equal to the element spacing $d = 75$ mm. The angular separation between elements ($\Delta\alpha$) from the cylinder center is determined as follows:

$$\Delta\alpha = d / R = 75 \text{ mm} / 300 \text{ mm} = 0.25 \text{ radians}$$

Converting the angular spacing from radians to electrical degrees:

$$\Delta\alpha = 0.25 \times (180 / \pi) \approx 14.32^\circ$$

Due to the structural symmetry of the 1×4 linear array about the broadside centerline axis (0°), the exact angular positions of the individual elements are:

- Inner Elements (Ports 2 & 3): $\pm \Delta\alpha / 2 = \pm 7.16^\circ$
- Outer Elements (Ports 1 & 4): $\pm 3\Delta\alpha / 2 = \pm 21.48^\circ$

2.2 Spatial Path Delay Projection (Δz)

By establishing the spatial phase center reference plane at the inner elements ($z = 0$ for Ports 2 and 3), the outer elements (Ports 1 and 4) are physically set back along the main z-axis direction. The projection distance (Δz) is calculated via trigonometry:

$$\Delta z = R \times [\cos(\Delta\alpha / 2) - \cos(3\Delta\alpha / 2)]$$

$$\Delta z = 300 \text{ mm} \times [\cos(7.16^\circ) - \cos(21.48^\circ)]$$

$$\Delta z = 300 \text{ mm} \times [0.9922 - 0.9306] = 300 \text{ mm} \times 0.0616 \approx 18.48 \text{ mm}$$

2.3 Phase Compensation Calibration

To mathematically flatten the curved wavefront and restore broadside radiation functionality, a phase lead must be applied to the outer ports to compensate for the 18.48 mm physical retreat path delay:

$$\Phi_{\text{correction}} = (2\pi / \lambda_0) \times \Delta z = (360^\circ / 150 \text{ mm}) \times 18.48 \text{ mm} \approx 44.35^\circ$$

Calibrated Phase Correction Values for Broadside Alignment (0°):

- Port 1 (Outer Left): +44.4°
- Port 2 (Inner Left): 0.0°
- Port 3 (Inner Right): 0.0°
- Port 4 (Outer Right): +44.4°

3. Phased Array Progressive Scanning Calculations

To steer the main beam to an explicit target scanning angle (φ_s), the structural phase alignment values must be superimposed with the progressive linear path phase variations. The generalized phase distribution formula for an element at coordinates (x_n, z_n) is defined by:

$$\Phi_n = \beta \times [x_n \sin(\varphi_s) + z_n \cos(\varphi_s)]$$

Where β represents the free-space phase constant:

$$\beta = 360^\circ / \lambda_0 = 360^\circ / 150 \text{ mm} = 2.4^\circ / \text{mm}$$

3.1 Spatial Element Coordinates

With the geometric center of the cylinder configured as the absolute spatial coordinate origin (0,0), the exact physical locations of the element phase centers for $R = 300 \text{ mm}$ are:

- Port 1 (Outer Left): $x_1 = -109.86 \text{ mm}$, $z_1 = 279.17 \text{ mm}$
- Port 2 (Inner Left): $x_2 = -37.39 \text{ mm}$, $z_2 = 297.66 \text{ mm}$
- Port 3 (Inner Right): $x_3 = +37.39 \text{ mm}$, $z_3 = 297.66 \text{ mm}$
- Port 4 (Outer Right): $x_4 = +109.86 \text{ mm}$, $z_4 = 279.17 \text{ mm}$

3.2 Consolidated Phased Scanning Lookup Table

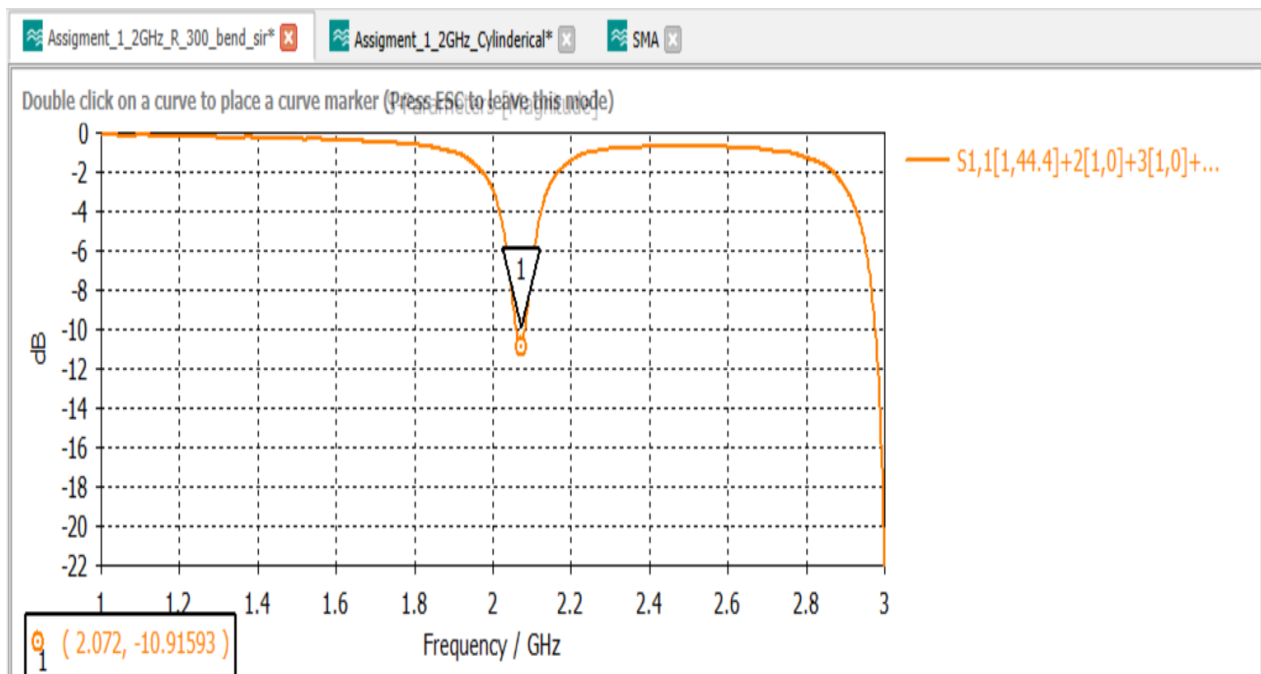
The following matrix summarizes the exact computed phase distributions utilized during CST Studio

electromagnetic co-simulation runs to steer the beam from 0° to 50°:

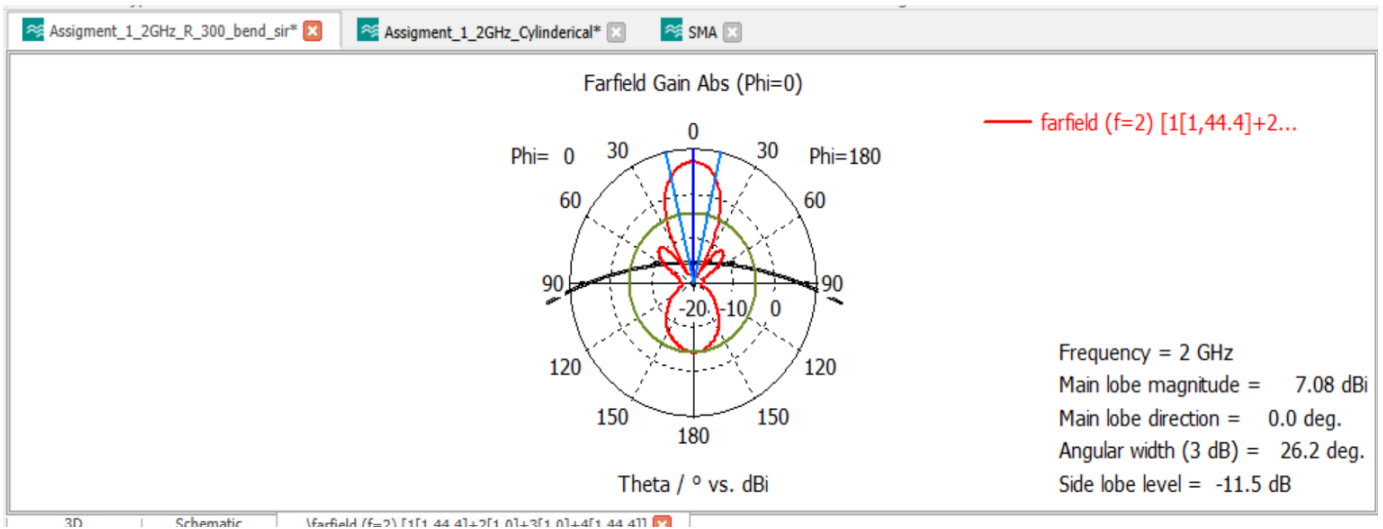
Scan Target Angle

Scan Target Angle (ϕ_s)	Port 1 Phase ($^\circ$)	Port 2 Phase ($^\circ$)	Port 3 Phase ($^\circ$)	Port 4 Phase ($^\circ$)
0° (Broadside)	+44.4°	0.0°	0.0°	+44.4°
10°	+44.4°	-31.3°	-62.5°	-49.4°
20°	+44.4°	-61.6°	-123.1°	-140.3°
30°	+44.4°	-90.0°	-180.0°	-225.6°
40° (Optimized)	+44.4°	-135.0°	-250.0°	-310.0°
50° (Optimized)	+44.4°	-160.0°	-295.0°	-375.0°

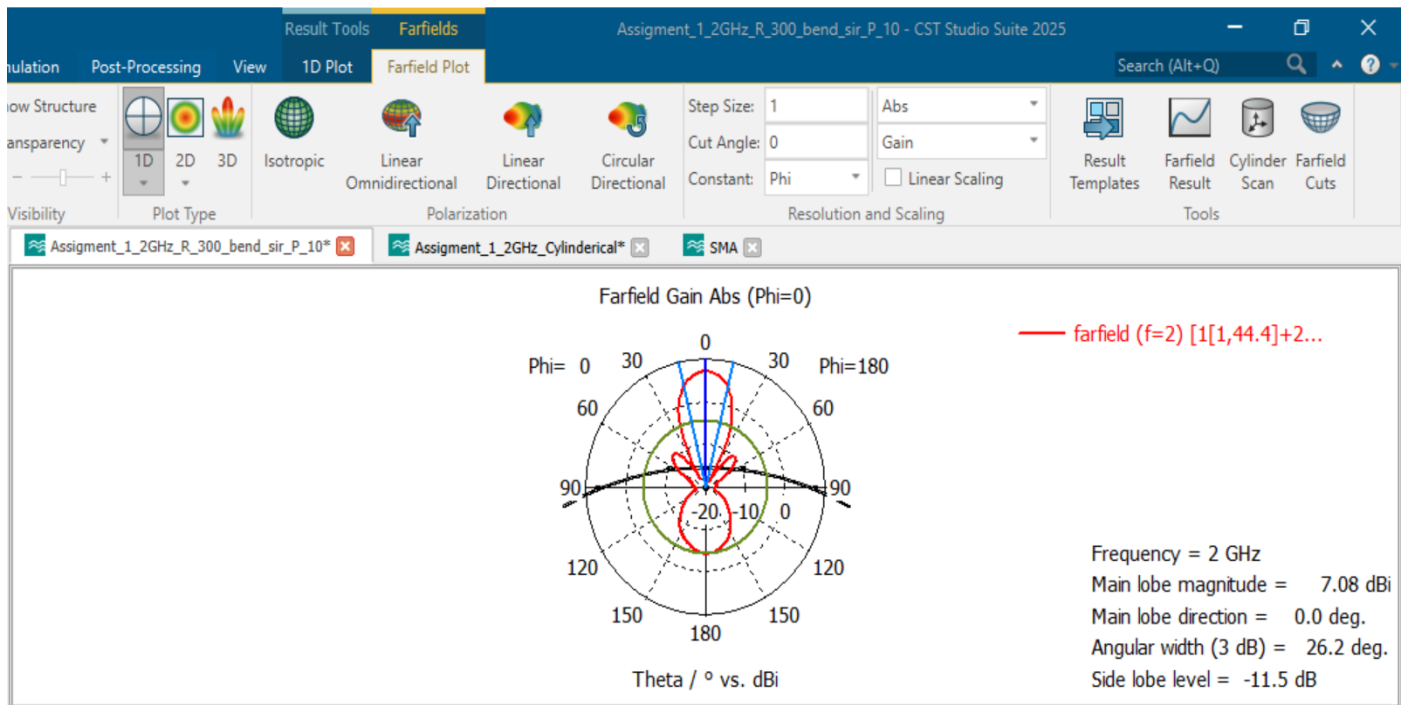
S11 Combine:



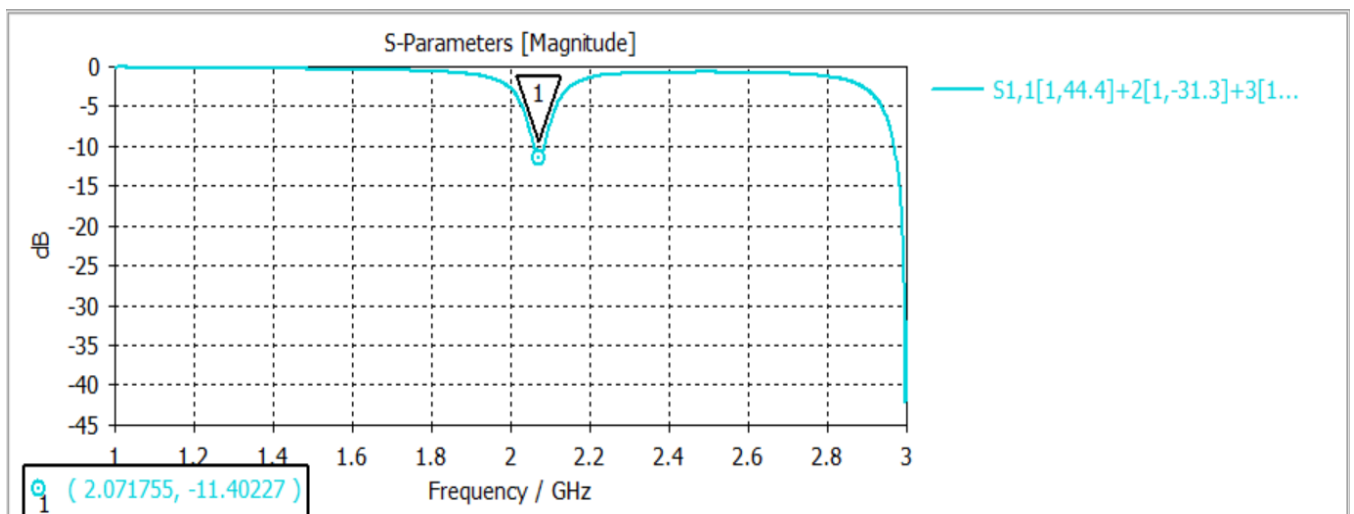
Gain:



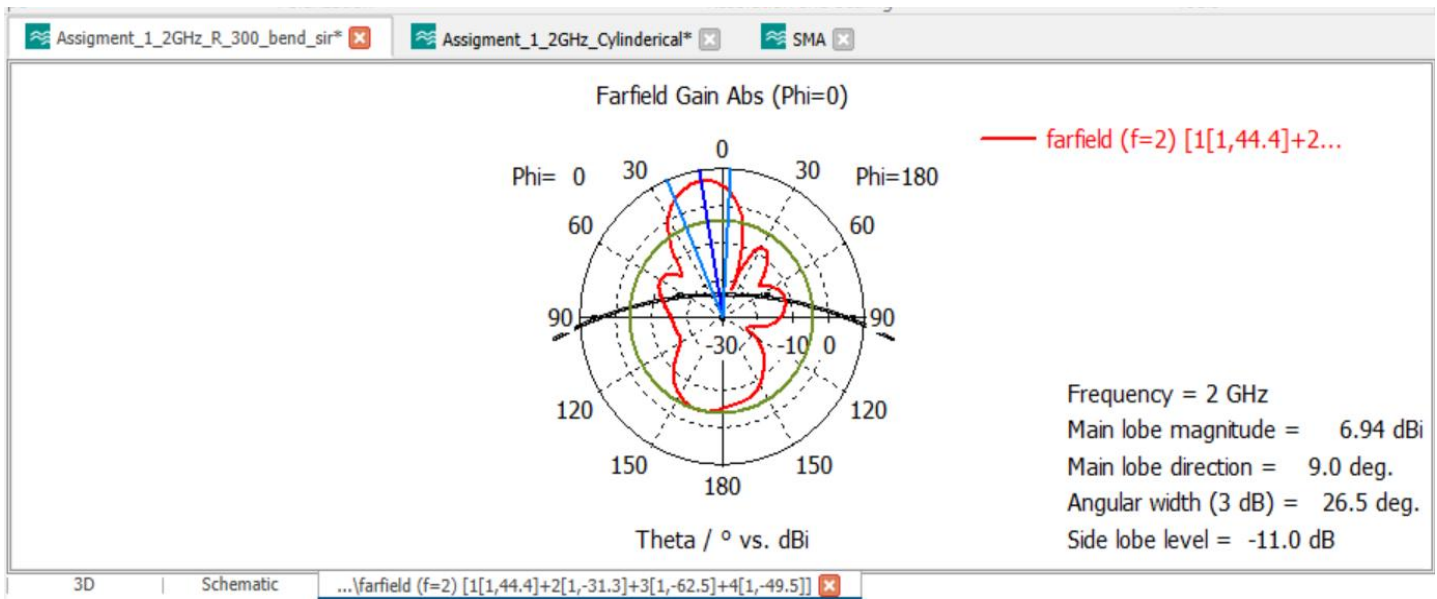
Realize Gain:



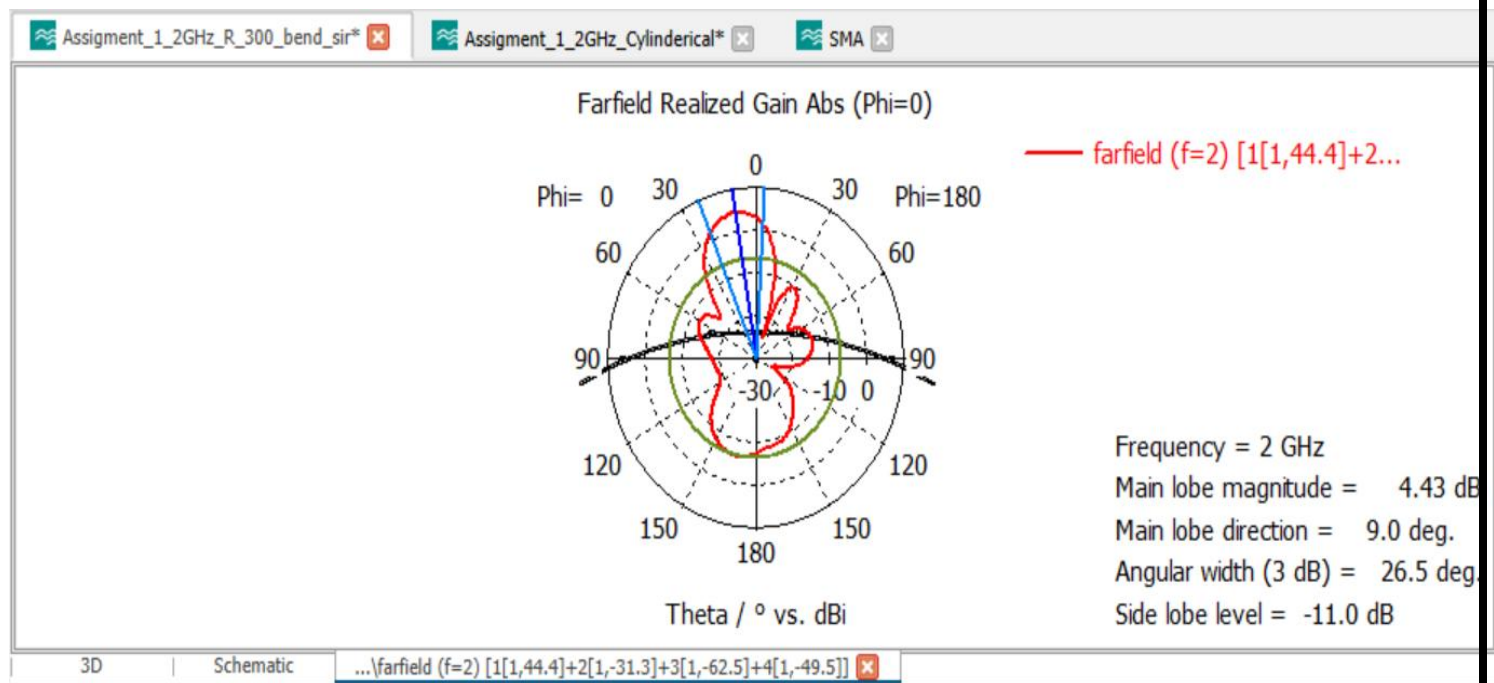
Phase 10:



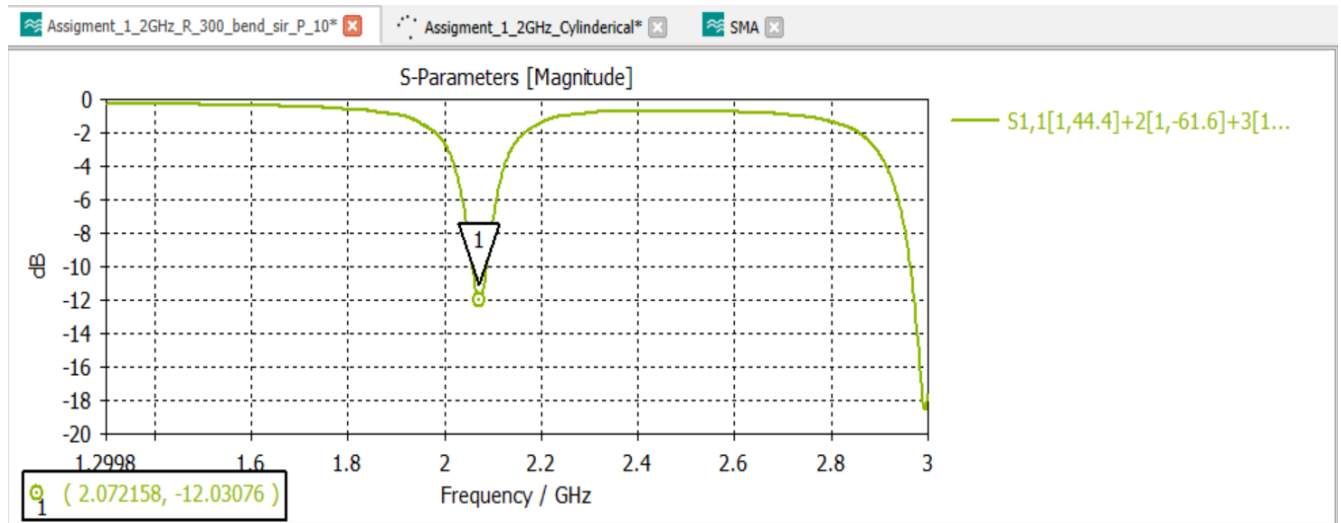
Gain:



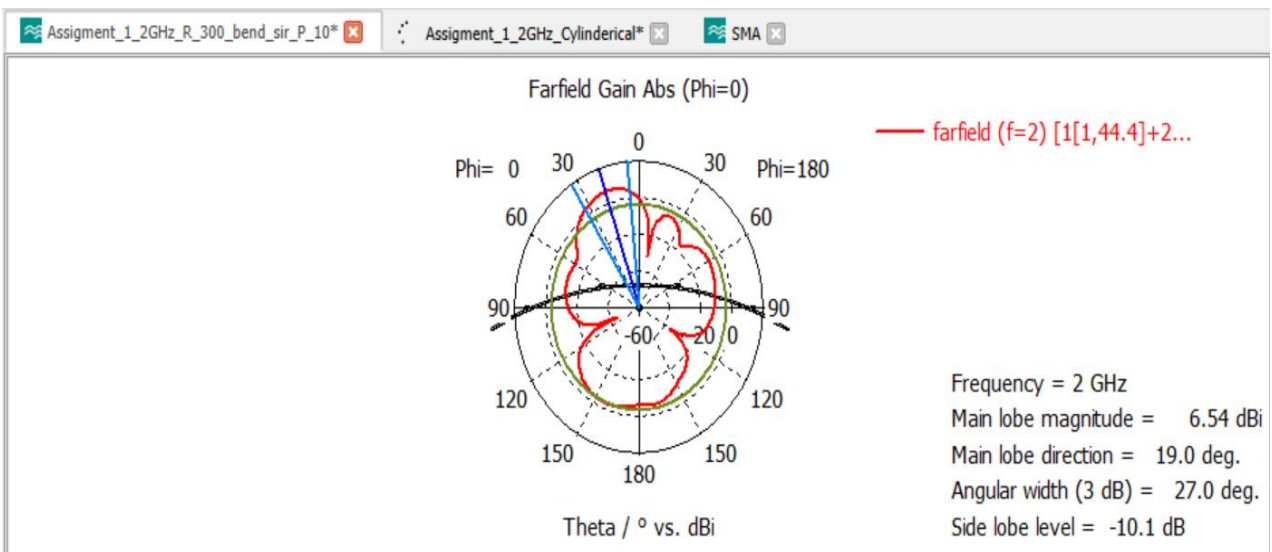
Realize Gain:



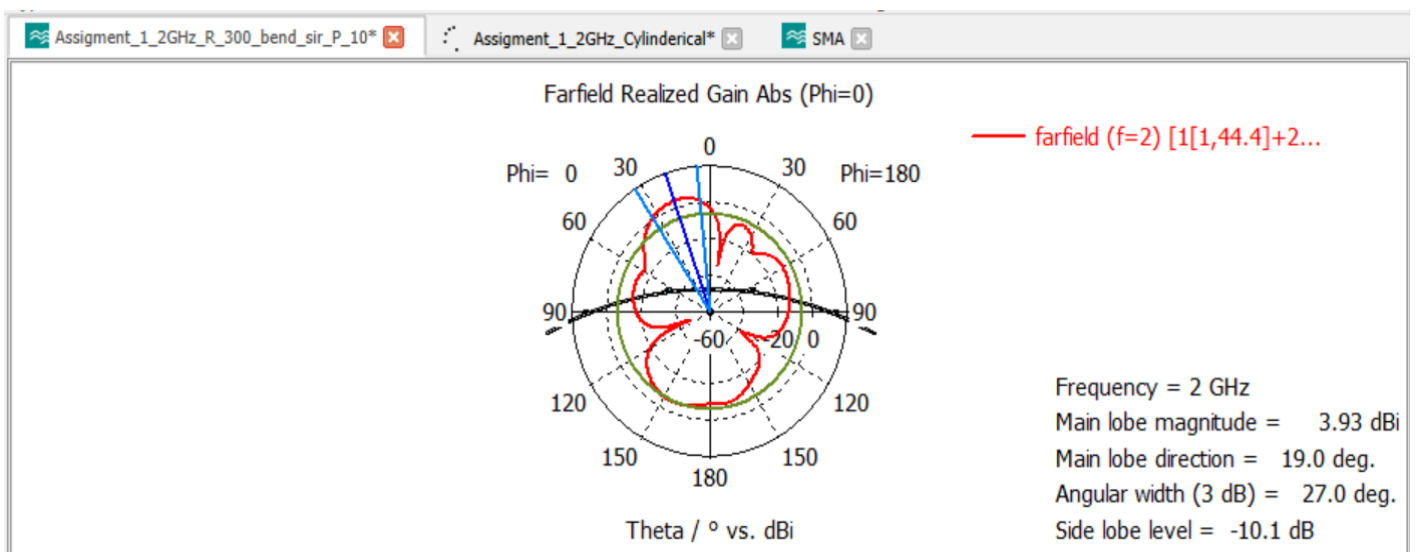
Phase 20:



Gain:

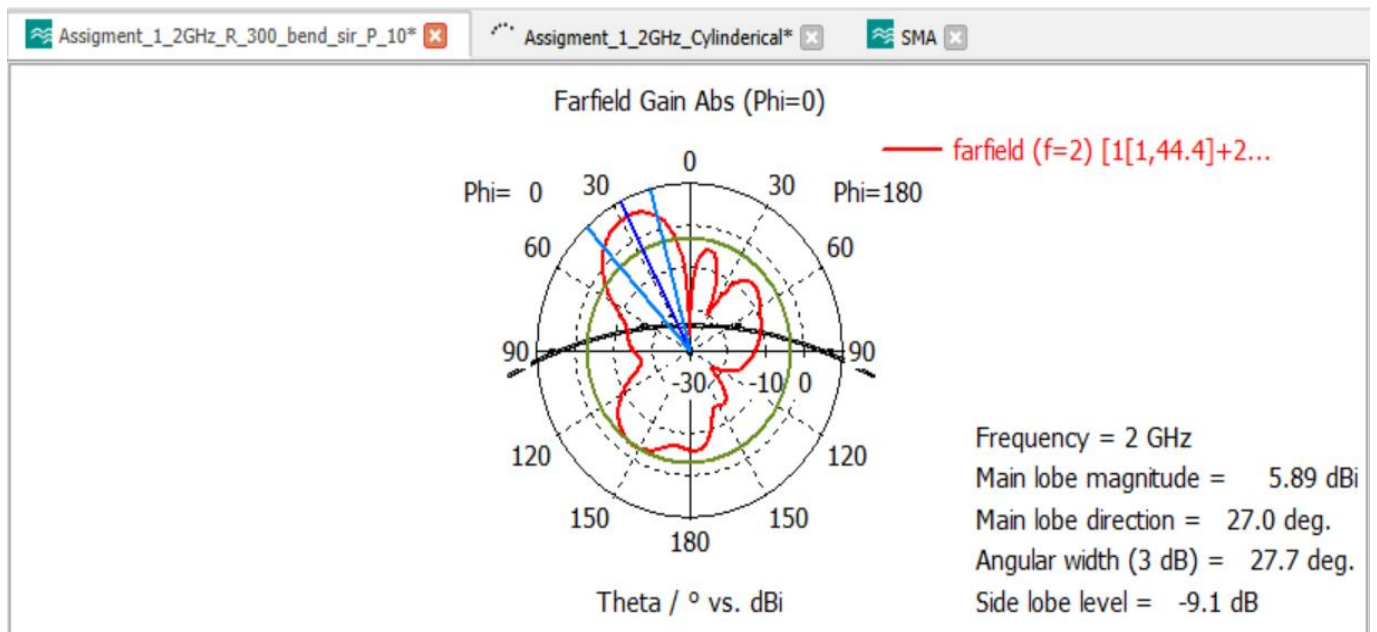


Realize Gain:

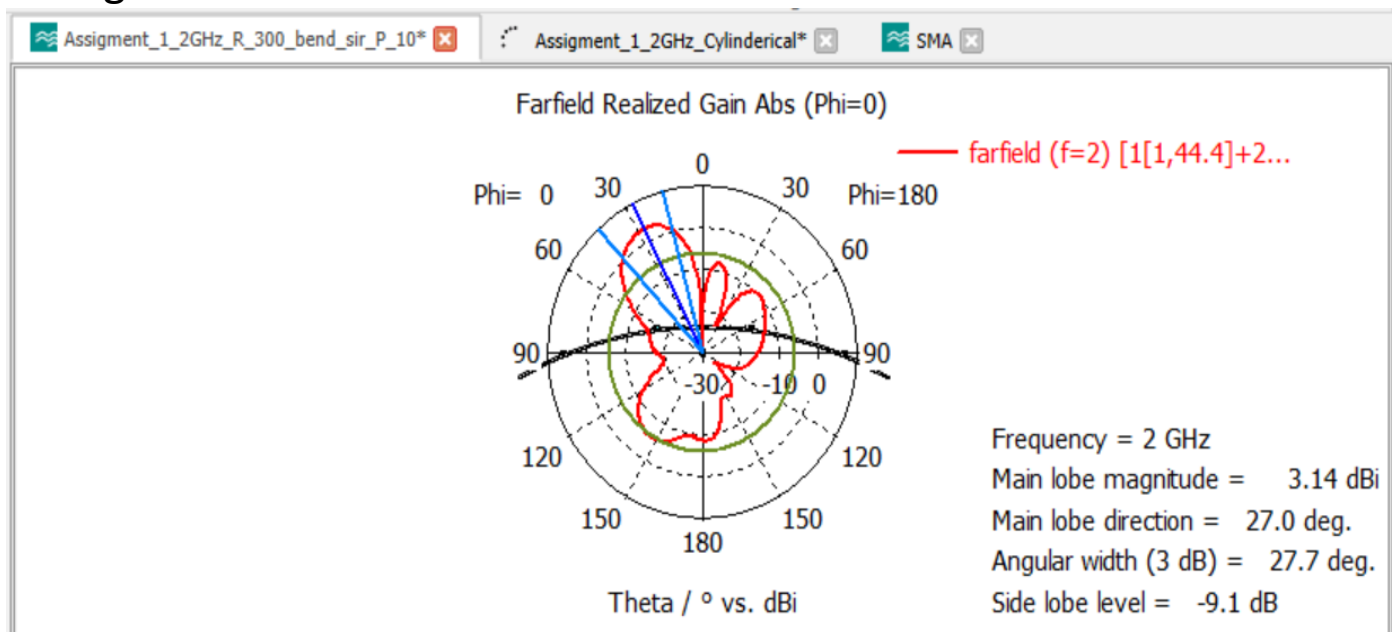


Phase 30:

Gain:

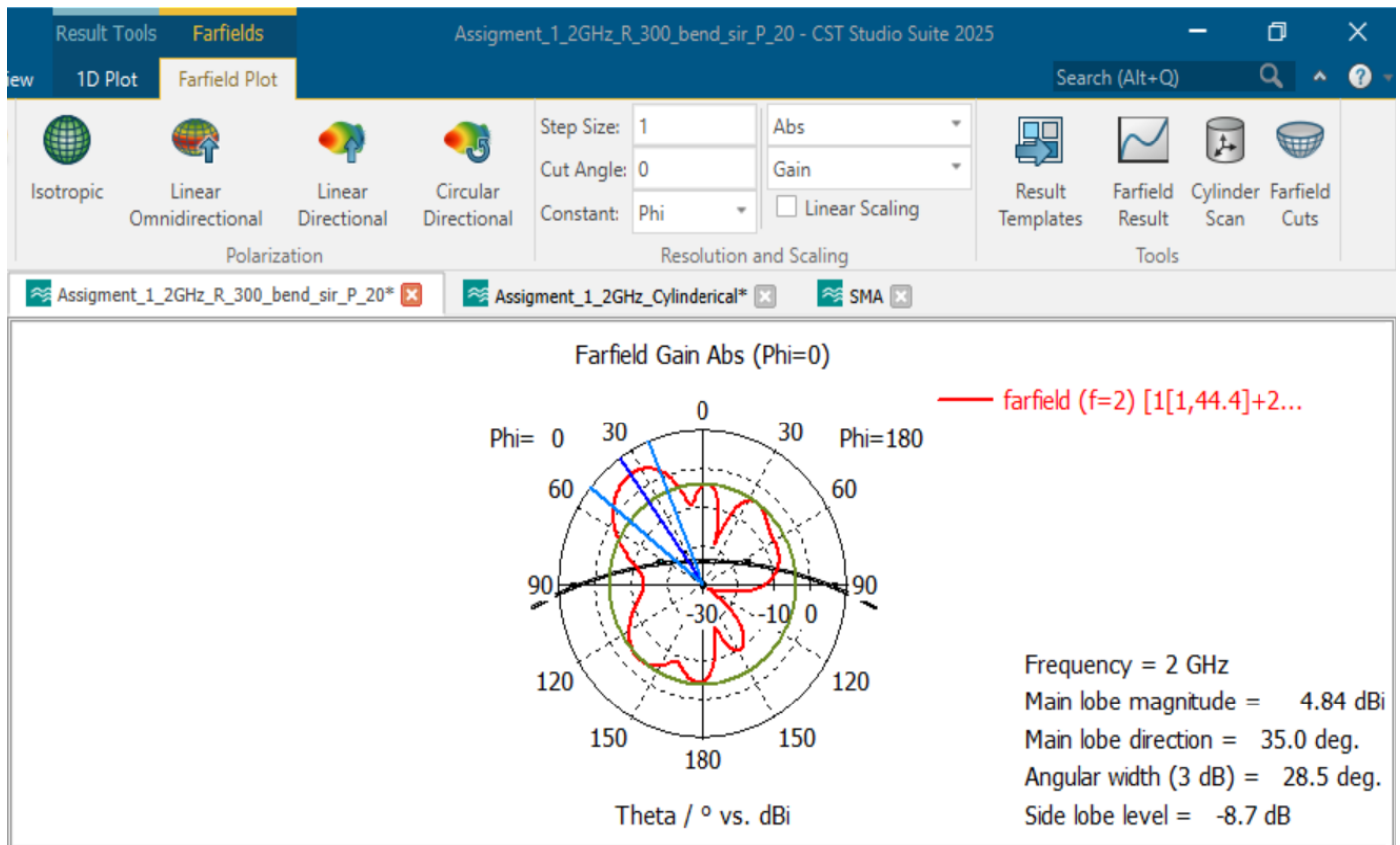


Realize gain:

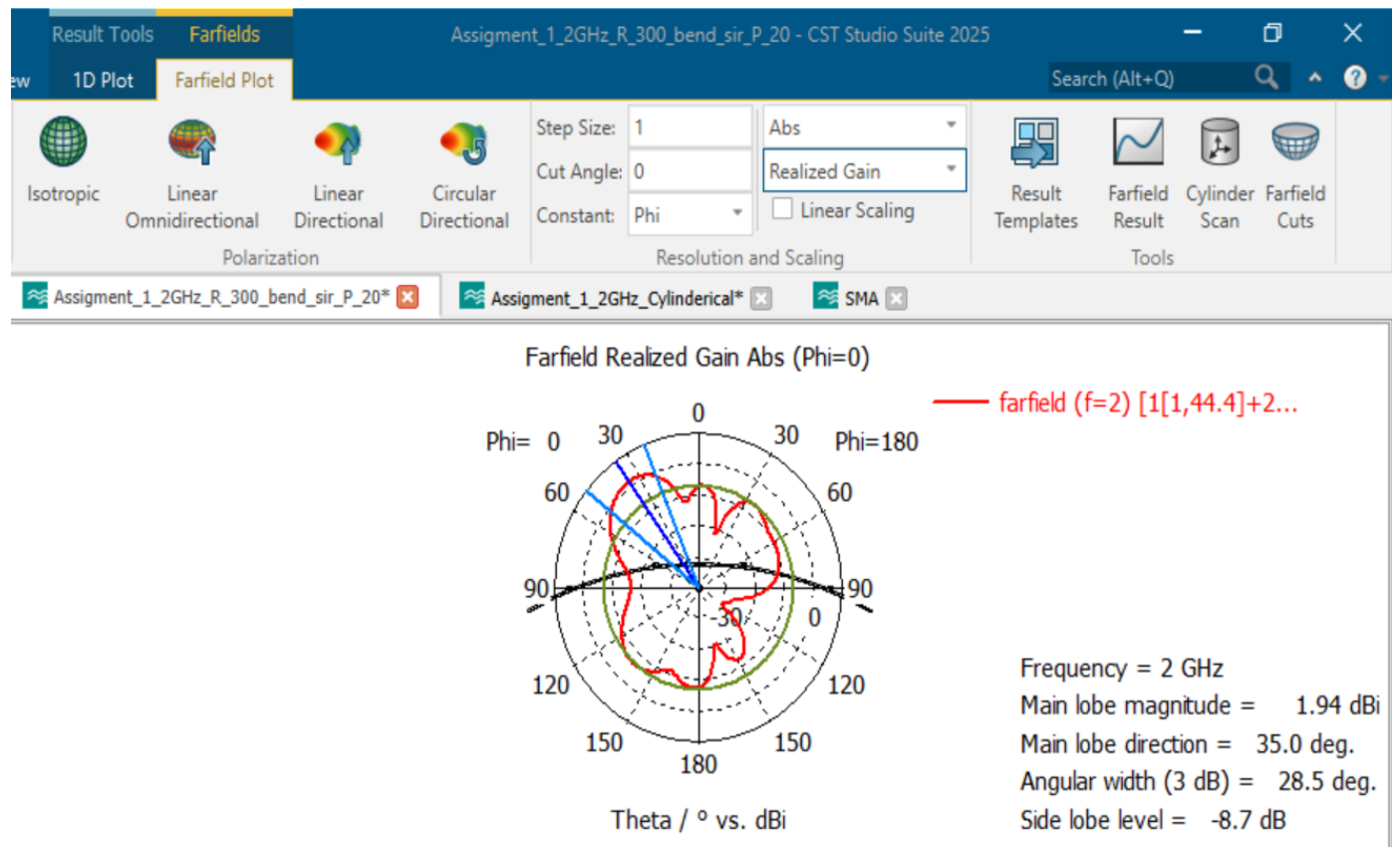


Phase 40:

Gain:

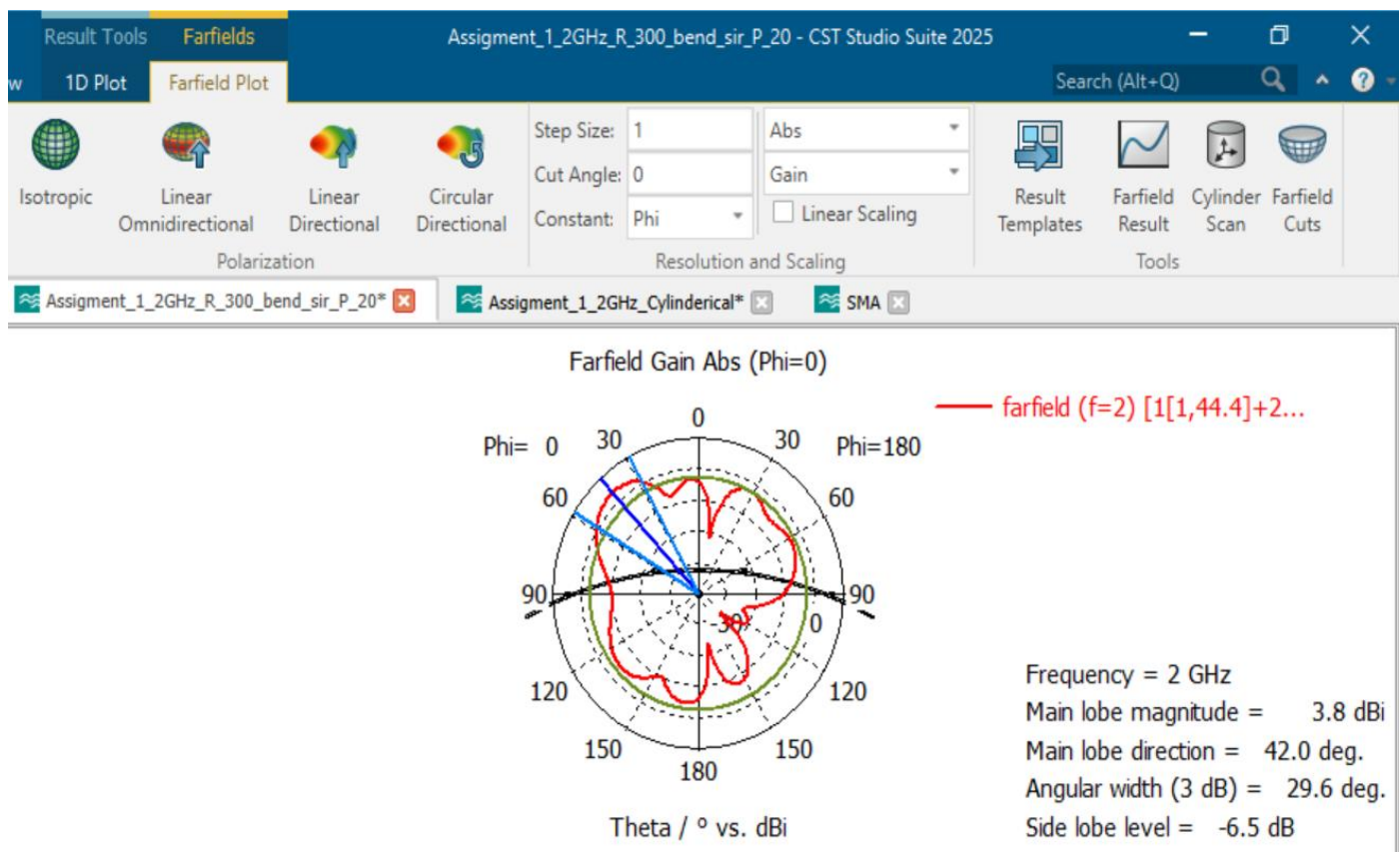


Realize Gain:



For Phase 50:

Gain:



Realize Gain:

